Exp No:12

Experiment to detect outliers in a given data set.

AIM:

Understand the procedure to identify the outliers in a given dataset

PROGRAM:

```
import numpy as np
array=np.random.randint(1,100,16)
array
array.mean()
np.percentile(array,25)
np.percentile(array,50)
np.percentile(array,75)
np.percentile(array,100)

def outDetection(array):
sorted(array)
Q1,Q3=np.percentile(array,[25,75])
IQR=Q3-Q1
lr=Q1-(1.5*IQR)
ur=Q3+(1.5*IQR)
return lr,ur
lr,ur=outDetection(array)

import seaborn as sns
%matplotlib inline
sns.displot(array)
sns.distplot(array)
new_array=array[(array>lr) & (array<ur)]
new_array

sns.displot(new_array)
lr1,ur1=outDetection(new_array)
lr1,ur1

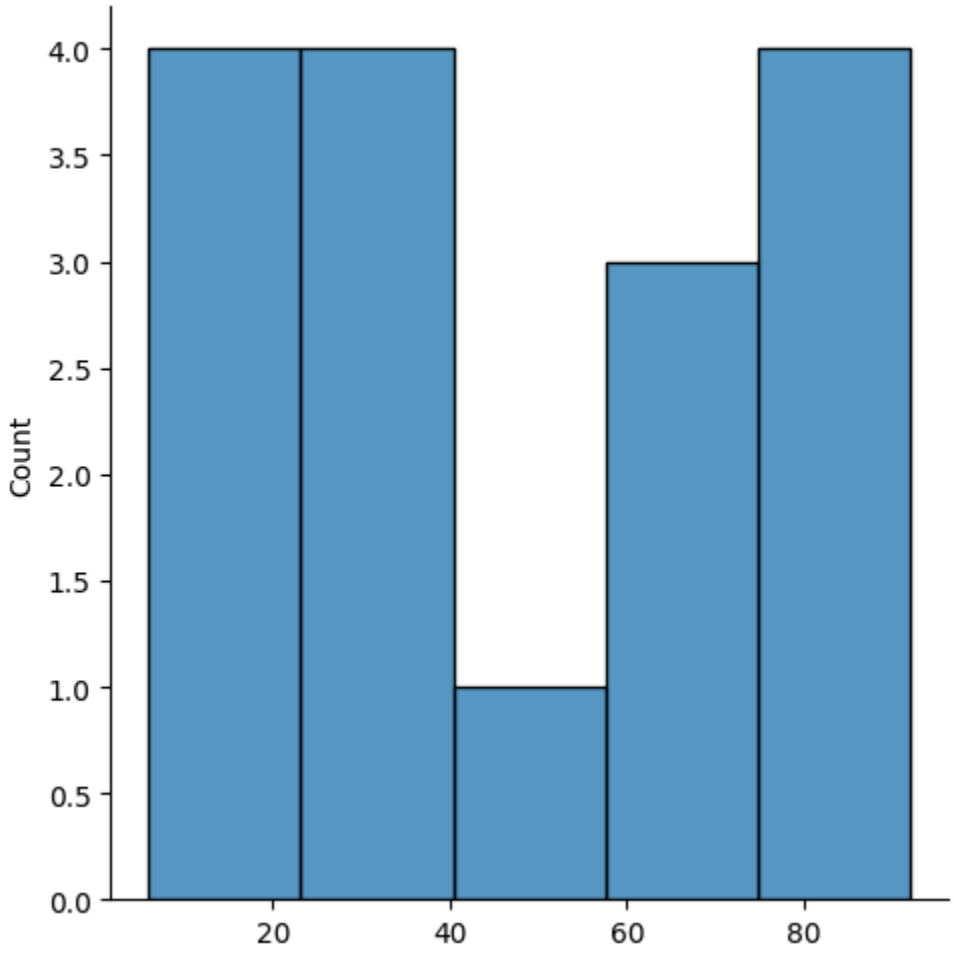
final_array=new_array[(new_array>lr1) & (new_array<ur1)]
Final_array
```

OUTPUT:

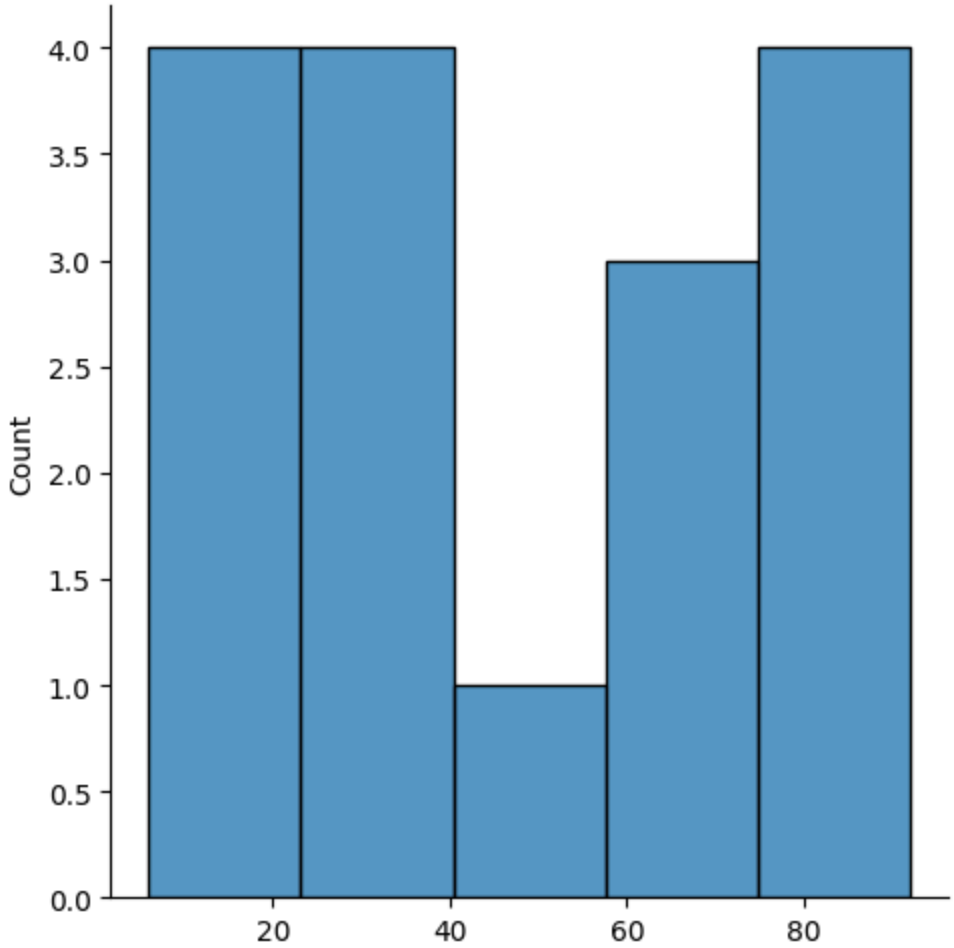
array([20, 23, 81, 25, 77, 92, 91, 34, 32, 61, 62, 20, 6, 55, 70, 24])
np.float64(48.3125)
np.float64(23.75)

np.float64(44.5)
np.float64(71.75)
np.float64(92.0)

(np.float64(-48.25), np.float64(143.75))
<seaborn.axisgrid.FacetGrid at 0x79deab491010>



array([20, 23, 81, 25, 77, 92, 91, 34, 32, 61, 62, 20, 6, 55, 70, 24])
<seaborn.axisgrid.FacetGrid at 0x79deab3c5bd0>



(np.float64(-48.25), np.float64(143.75))
array([20, 23, 81, 25, 77, 92, 91, 34, 32, 61, 62, 20, 6, 55, 70, 24])
/tmp/ipykernel_4353/209491988.py:1: UserWarning:

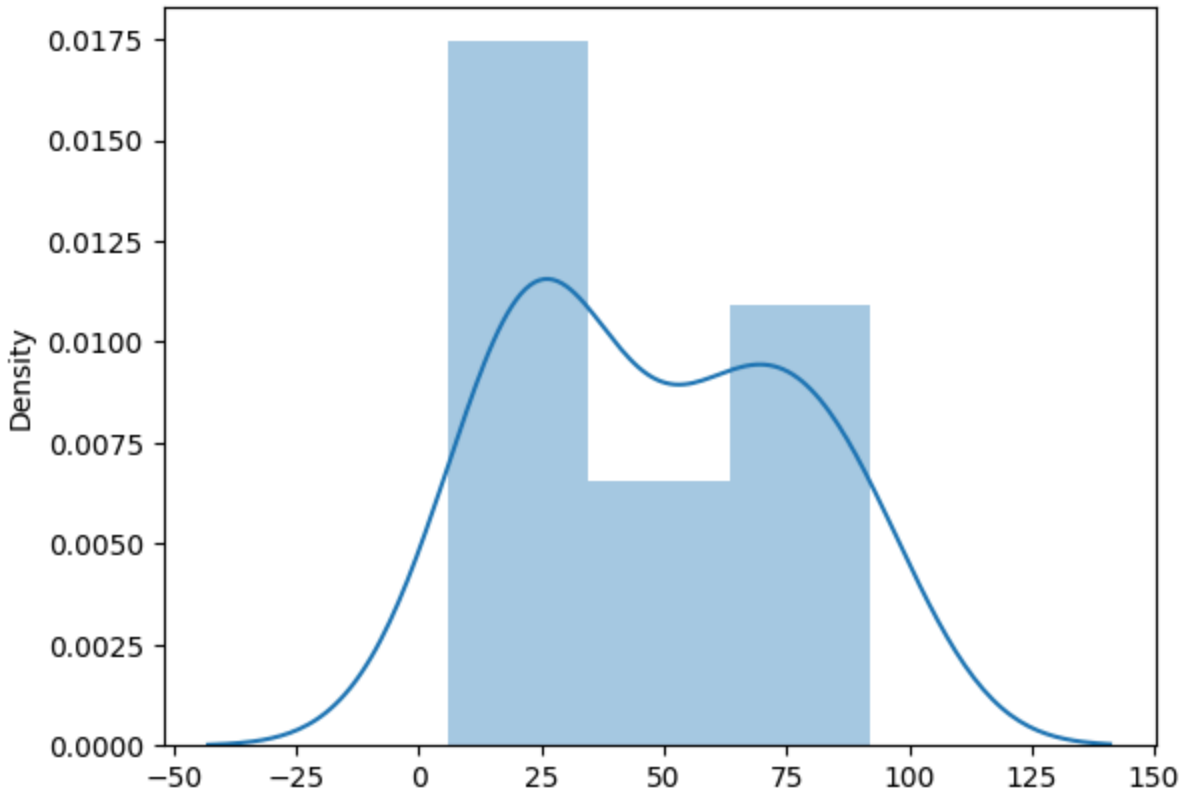
`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

```
sns.distplot(final_array)
```

<Axes: ylabel='Density'>



RESULT:

The experiment has been carried out successfully